

Perceptual and Acoustic Correlates of Racial Identity in Text-to-Speech Voices

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BACKGROUND

- Listeners recover **race** from voice alone: American English listeners reliably distinguish White and Black speakers from phonetic cues of Mainstream American English (MAE) vs. African American English (AAE) (Purnell et al. 1999, Thomas & Reaser 2004) — and judge voices heard as ‘sounding Black’ **more negatively**.
- Voice-AI is increasingly human-like and deployed in high-stakes settings, but findings on race in TTS are **mixed**: listeners converge on race labels (Holliday 2023), yet show a bias towards rating synthetic voices as ‘White’ (Pinhanez et al. 2024).

Research questions

- Q1 Can listeners perceive race in **fully synthetic** voices they know are synthetic?
- Q2 Does perceived race interact with **personality** judgments, and with perceived **human-likeness**?
- Q3 Which **acoustic features** distinguish Black- from White-rated voices?

METHODS

Perception experiment

- 144** American English listeners (Prolific), ethnically representative sample; **32 TTS voices** (EasyPeasy AI): 8 per platform-assigned label — {Black, White} × {Male, Female}. Audio only, amplitude-normalized to 72 dB.
- Personality block**: 16 voices read the Rainbow Passage; 7-point ratings for *friendly, professional, competent, trustworthy, pleasant, funny* + **human-likeness** (Holliday 2023).
- Race block**: the other 16 voices; 4 sentences targeting AAE-relevant vowels (/æ, ε, ι, i, a, ei, ai/), avoiding AAE morphosyntactic cues; forced-choice race (Asian, Black, Hispanic/Latino, White) + age, region, gender; 64 trials.

Acoustic features

(Psychoacoustic Model of the Voice; Kreiman et al. 2014)

- Per vowel token, at 1-ms frames (VoiceSauce; Shue et al. 2009): F0, **F1–F4**, formant dispersion; **Residual H1***, H2*–H4*, H4*–H2kHz*, H2kHz*–H5kHz*, **CPP**, SHR, Energy.
- MAD outlier exclusion; min–max normalization within gender; **mean** + **CoV** at vowel onset/midpoint/offset; centered by phonation modality.

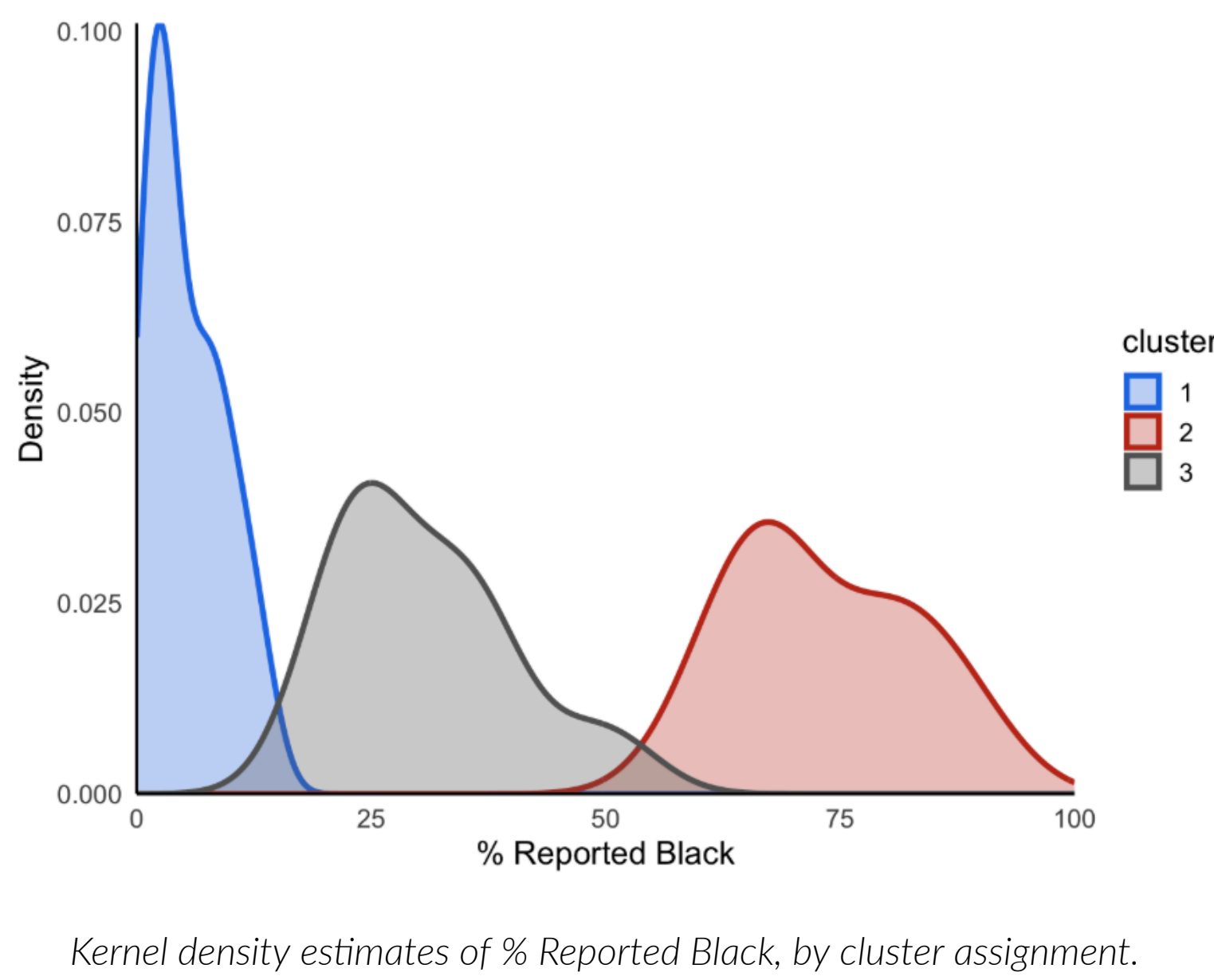
Classifier

- Gradient Boosted Decision Trees** (XGBoost; Chen et al. 2024) classify Black- vs. White-rated voices from 61 features; speaker-level 5-fold cross-validation × 50 repeats, grid-searched hyper-parameters.

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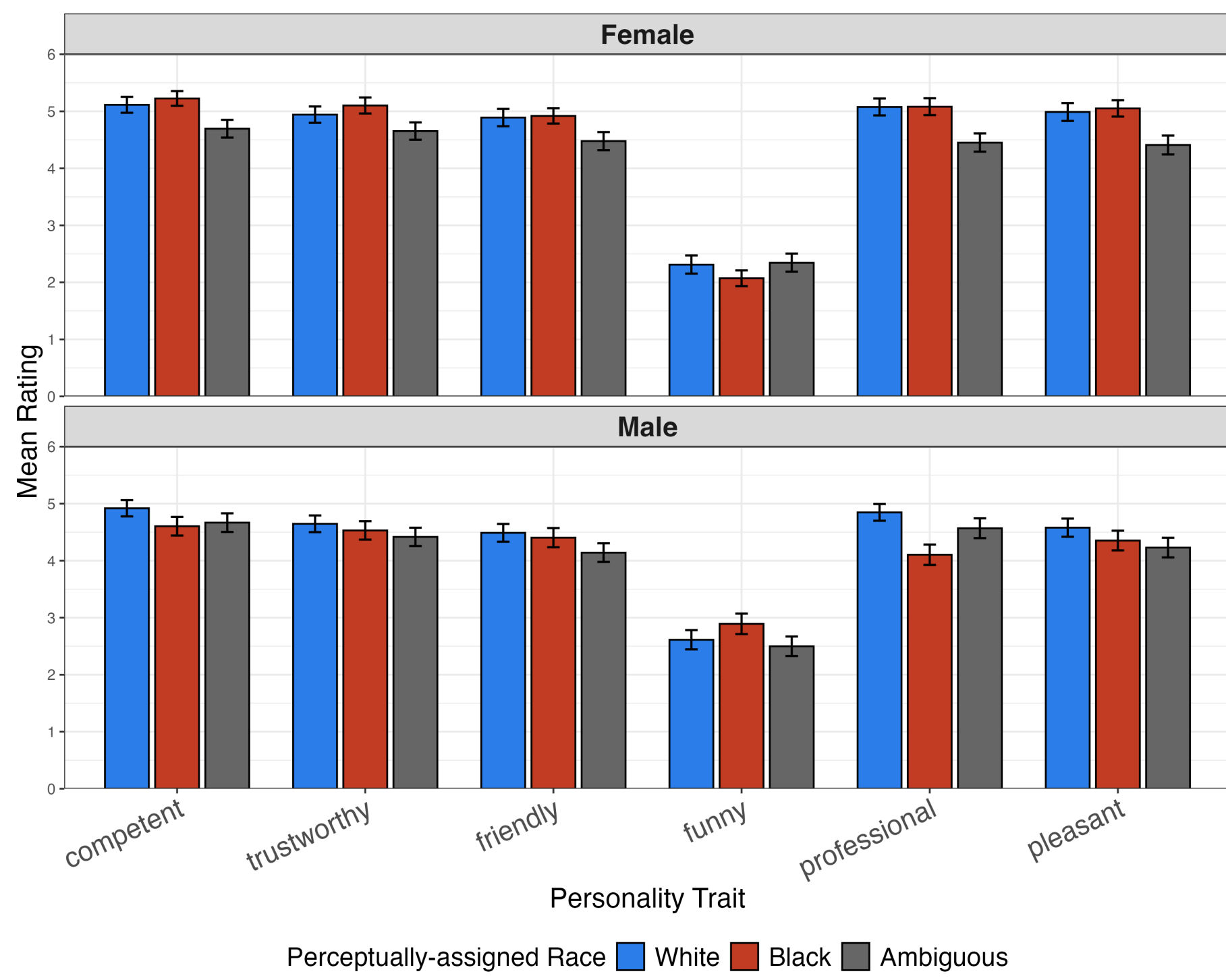
PERCEPTUAL RESULTS: RACE

- Agreement with platform-assigned **gender** was near ceiling (Male: 98.6%; Female: 93.6%).
- Perceptually grounded **race labels** derived by k-means clustering on % Reported Black (RB): voices consistently rated **White** (~0–20% RB), **Ambiguous** (~20–50%), or **Black** (~50–100%).



- The Black-rated cluster is much **wider** → more rater variability, and an overall **bias towards rating voices as White**.
- 5/9** Ambiguous voices were platform-labeled *Black Female*: Black female voices are harder to identify, as with human voices (Thomas & Reaser 2004).

PERCEPTUAL RESULTS: PERSONALITY

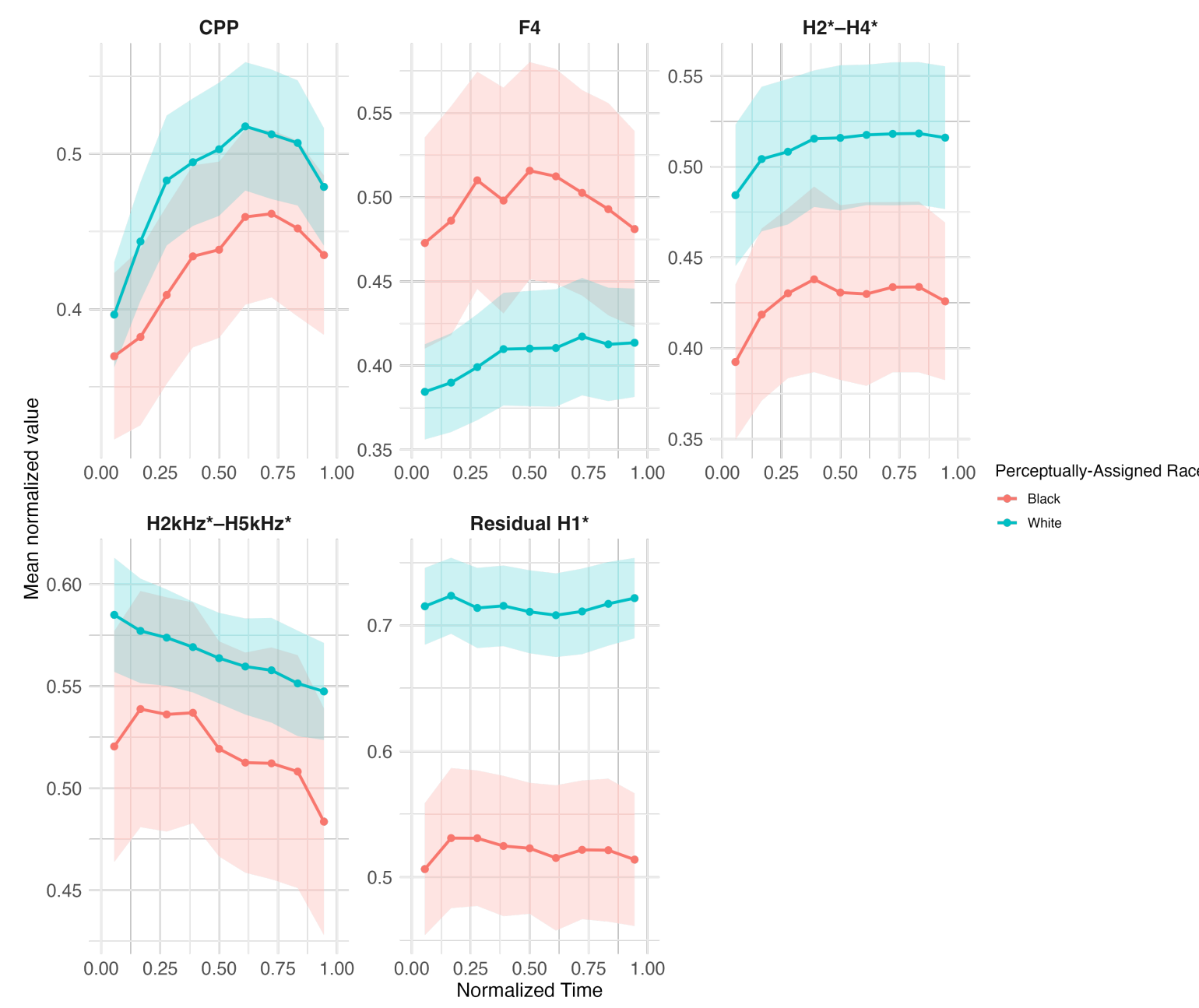


Mean personality ratings by perceptually-assigned race and gender.

- Per-trait linear mixed-effects models: perceived race + participant ethnicity + human-likeness; random intercepts for participant and voice; fit separately by perceived gender.
- Female voices**: no effect of perceived race on any trait. **Male voices rated Black** were rated as less *pleasant* ($\beta = -0.64$), *professional* ($\beta = -1.09$), *trustworthy* ($\beta = -0.53$), and *competent* ($\beta = -0.67$), all $p < .05$ → **racial stereotyping of synthetic voices**.
- Human-likeness** positively predicted **all** traits for both genders ($p < .01$); participant ethnicity was never significant.

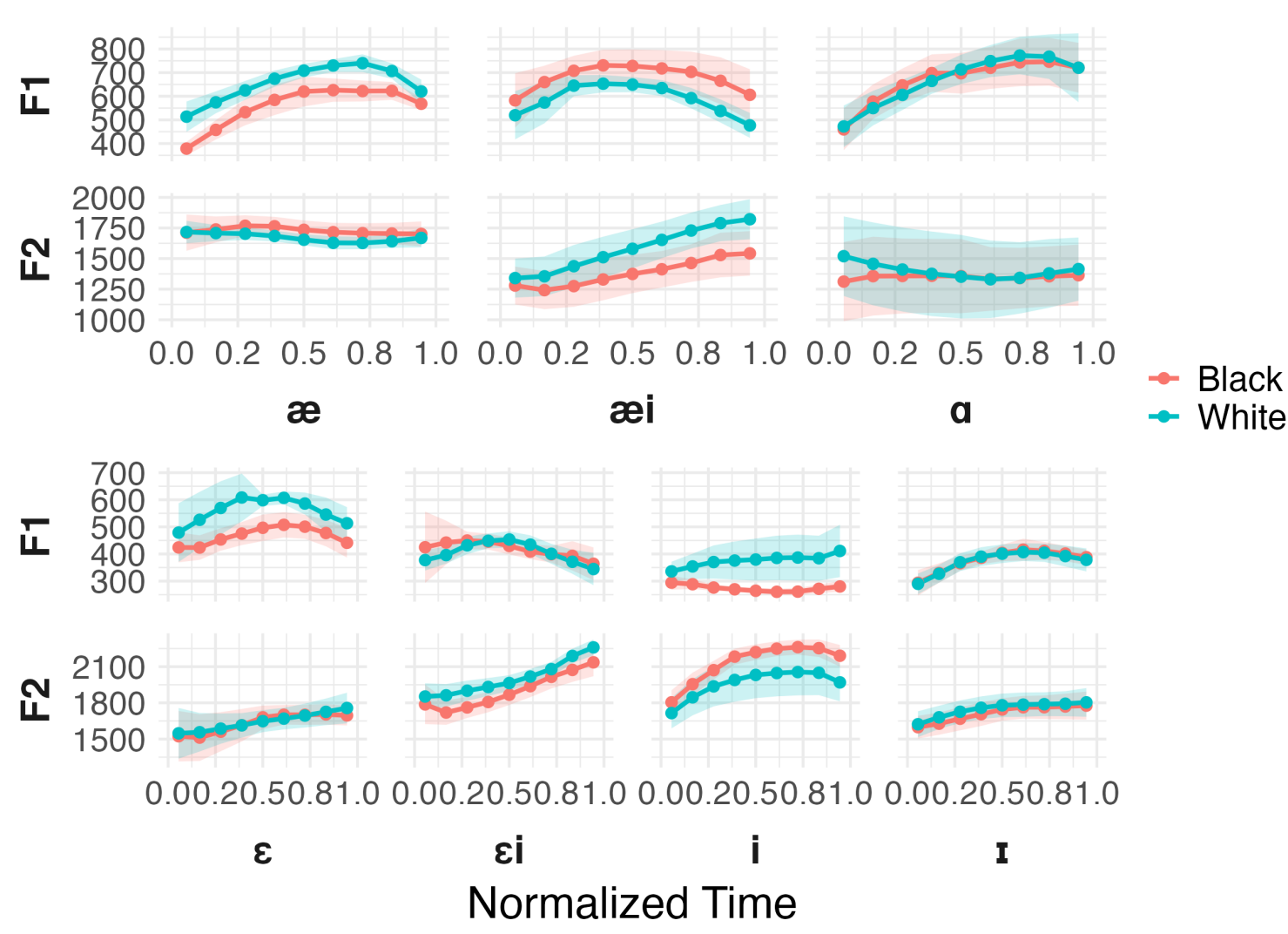
ACOUSTIC RESULTS

- Ambiguous voices excluded; classifier trained on **male-rated** voices (too few Black Female voices after filtering). Full model: **65%** CV accuracy; reduced top-15-feature model: **66%**.
- Black-rated voices** show lower **CPP**, **H2*–H4***, and **H2kHz*–H5kHz***, higher **F4**, and markedly lower **Residual H1*** — the most robust effect, linked to glottal constriction (Chai & Garellek 2022).



Mean structure across normalized time for the top classifier features (red = Black-rated, teal = White-rated); ribbons = 95% CI.

- Vowel quality**: localized ~100 Hz F1 differences for /æ, ε, i/, and F1/F2 differences for /ai/ — mirroring documented AAE–MAE vowel differences in human speech (Kohn 2013).



Raw F1/F2 trajectories for Black- vs. White-rated voices; ribbons = 95% CI.

DISCUSSION

- (Q1) Listeners **can** perceive race in fully synthetic speech, though with a bias towards ‘White’ ratings.
- (Q2) Listeners **still stereotype**: male voices rated Black receive lower personality ratings, and human-likeness boosts all personality attributions (anthropomorphism) (Schreibelmayr & Mara 2022).
- (Q3) Cues tied to laryngeal configuration (**Residual H1***, H2*–H4*), speaker individuality (CPP, F4; Lee et al. 2019), and **AAE-like vowel quality** distinguish Black- from White-rated voices.
- Implications**: these acoustics can guide the design of racially-marked voices, but stereotyping warrants **caution** when deploying synthetic speech in socially sensitive contexts.